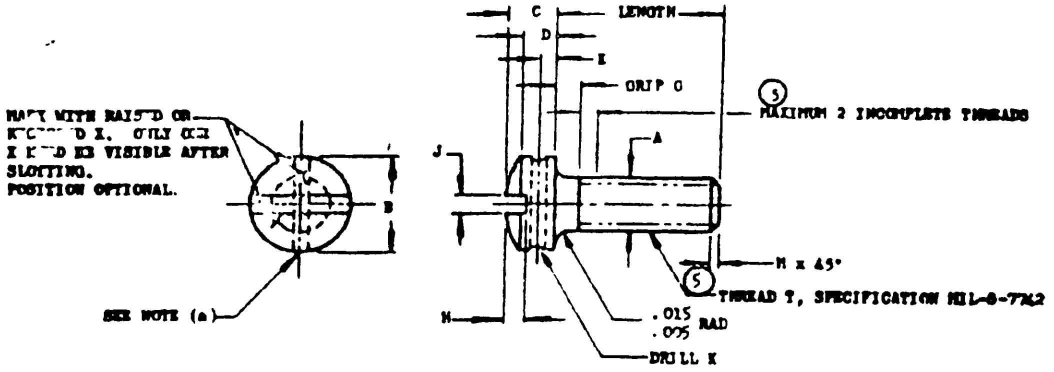
NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all conflicting standard drawings for the same product and shall become effective for the procurement of aeronautical supplies, or for use in new design, not later than 6 months after the latest date of approval shown.


MARK WITH RAISED OR RECESSED X. ONLY ONE X NEED BE VISIBLE AFTER SLOTTING. POSITION OPTIONAL.
SEE NOTE (a)
B
J
H
C
D
E
LENGTH
GRIP G
⑤ MAXIMUM 2 INCOMPLETE THREADS
A
M x 45°
⑤ THREAD T, SPECIFICATION MIL-S-7742
.015 .005 RAD
DRILL K

⑤ THREAD T	A MAX DIA	⑤ B DIA		C		D		(c) E REF	GRIP G		H		J		⑤ K DRILL +.004 −.001	M MAX
		MAX	MIN	MAX	MIN	MAX	MIN		MAX	MIN	MAX	MIN	MAX	MIN		
NO. 6-32 UNC-3A	.1380	.226	.208	.132	.118	.096	.089	.043	.063	.031	.045	.035	.048	.039	.046	.031
NO. 8-32 UNC-3A	.1640	.270	.250	.156	.141	.113	.106	.043	.063	.031	.065	.054	.056	.045	.046	.031
(d) NO. 10-24 UNC-3A	.1900	.313	.292	.180	.165	.130	.123	.043	.063	.031	.075	.064	.060	.050	.046	.031
1/4-20 UNC-3A	.2500	.414	.389	.237	.219	.170	.161	.062	.078	.046	.102	.087	.075	.064	.048	.031
5/16-18 UNC-3A	.3125	.518	.490	.295	.276	.211	.201	.078	.078	.046	.130	.110	.084	.072	.070	.047

LENGTH		DASH NUMBERS				
		NO. 6-32	NO. 8-32	(d) NO. 10-24	1/4-20	5/16-18
1/8 1/4	+0 −1/32	(b) ~~6-2~~ 6-4	(b) ~~8-2~~ 8-4	(b) ~~10-4~~		
3/8 1/2		6-6 6-8	8-6 8-8	10-6 10-8	416-8	
5/8 3/4			8-10 8-12	10-10 10-12	416-10 416-12	516-12
7/8 1			8-14 8-16	10-14 10-16	416-14 416-16	516-14 516-16
1-1/8 1-1/4	+0 −1/16			10-18 10-20	416-18 416-20	516-18 516-20
1-3/8 1-1/2				10-22 10-24	416-22 416-24	516-22 516-24
1-5/8 1-3/4				10-26	416-26 416-28	516-26 516-28
1-7/8 2				10-30	416-30 416-32	516-30 516-32

⑤ (a) DRILLED HOLE TO BE ON CENTER APPROXIMATELY AT RIGHT ANGLES TO THE SLOT BUT MUST NOT BREAK THROUGH THE BASE OR TOP (SPHERICAL PORTION) OF THE HEAD.

(b) PART NUMBERS AN503-6-2, AN503-8-2, AND AN503-10-4 INACTIVE FOR DESIGN AFTER 23 JANUARY 1946.

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

(c) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES ONLY AND ARE NOT AN INSPECTION REQUIREMENT.

⑤ (d) NOT PREFERRED FOR AIRFRAME USE.

MATERIAL: STEEL, SPECIFICATION AN-S-9, MIL-S-6050, OR OTHER STEEL MEETING THE SAME PHYSICAL REQUIREMENTS.

HEAT TREAT: 125,000 TO 145,000 PSI TENSILE STRENGTH, SPECIFICATION MIL-H-6875.

FINISH: CADMIUM PLATE, SPECIFICATION QQ-P-416, TYPE I, CLASS C.

EXAMPLE OF PART NO.: AN503-8-10 = NO. 8-32 SCREW, 5/8 INCH LONG

SCREWS SHALL BE FREE OF ALL LOOSE OR HANGING BURRS OR SLIVERS WHICH MIGHT BECOME DISLODGED UNDER USAGE.

DIMENSIONS IN INCHES.

APPROVED 12 Apr 44 REVISED ① 10 Jun 45 ② 23 Jan 46 ① 12 May 50 ④ 5 Jun 51 ⑤ 11 May 53

PROCUREMENT SPECIFICATION
⑤ MIL-S-7839

AIR FORCE-NAVY AERONAUTICAL STANDARD

SCREW, MACHINE, DRILLED FILLISTER
HEAD, COARSE THREAD

AN503

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